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Total No. of Questions : 5]

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B.Tech. IVth Semester (New Scheme)

Examination, 2023

Signal and Systems

Paper - EL-404

Time : 3 Hours]

[Maximum Marks : 60

Note :- Attempt all questions. Each questions carry equal marks.

1. Show that the product of two even signals or of two odd signals is an even signal and that the product of an even and an odd signal is an odd signal.

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(1)

P.T.O.

OR

Determine whether the following signals are energy signals, power signals or neither.

- (a) $x(t) = e^{-at} u(t)$, $a > 0$
- (b) $x(t) = t u(t)$
- (c) $x(n) = (-0.5)^n u(n)$

2. Perform circular convolution of the following two sequences.

$$x_1(n) = [1, 2, 2, 1] \text{ and } x_2(n) = [2, 1, 1, 2]$$

OR

Explain Parseval's Theorem and Central Ordinat Theorem with example.

3. Explain Laplace shifting properties. Find the Laplace Transform of the following functions:

- (a) $f(t) = \sin at \cos bt$
- (b) $f(t) = \frac{1-e^t}{1}$
- (c) $f(t) = \delta(t^2 - 3t + 2)$

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(2)

OR

Compute the inverse Laplace transform of

$$Y(s) = \frac{1}{3-4s} + \frac{3-2s}{s^2+49} \text{ and discuss its properties.}$$

4. Determine the inverse z-transform of $X(z) = \frac{z^2}{(z-a)^2}$ ROC

$|z| > |a|$ using contour integration-residue method.

OR

Find $x(n)$ by using convolution for

$$X(z) = \frac{1}{\left(1 - \frac{1}{2}z^{-1}\right)^2 \left(1 + \frac{1}{4}z^{-1}\right)^2}$$

5. Explain in detail cumulative Distribution Function (CDF) and Probability Density Function (PDF). In detail.

OR

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(3)

P.T.O.



Describe any two :

- (a) Wiener - Khinchin - Einstein Theorem
- (b) Central Limit Theorem
- (c) Auto-Correlation Function (ACF)
- (d) Cross Correlation Function (CCF)

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(4)

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